

Sequential Neural Score Estimation: Likelihood-Free Inference with Conditional Score Based Diffusion Models

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Simulation-Based Inference Problem

Problem Setup

Given a prior $p(\theta)$ and a simulator $p(x|\theta)$, infer posterior:

$$p(\theta|x_{obs}) = \frac{p(x_{obs}|\theta)p(\theta)}{p(x_{obs})}$$

Key Challenge

- ▶ Likelihood $p(x|\theta)$ is intractable
- ▶ Can sample from simulator but cannot evaluate density
- ▶ Standard MCMC and VI methods are not applicable

Applications

Neuroscience, evolutionary biology, epidemiology, climate science, cosmology, econometrics

Existing SBI Approaches

Neural Density Estimation Methods

- ▶ SNPE: Learn posterior $q_{\psi}(\theta|x) \approx p(\theta|x)$ using normalizing flows
- ▶ SNLE: Learn likelihood $q_{\psi}(x|\theta) \approx p(x|\theta)$, then use MCMC
- ▶ SNRE: Learn likelihood ratio $r(x, \theta) = p(\theta|x)/p(\theta)$

Limitations

- ▶ Normalizing flows require invertibility constraints
- ▶ GANs suffer from training instability
- ▶ Need for careful architectural design

Score-Based Diffusion Models: Key Idea

Score Function

Instead of modeling density $p(\theta|x)$, model its gradient:

$$s(\theta) = \nabla_{\theta} \log p(\theta|x)$$

Advantages

- ▶ No normalization constant needed
- ▶ Architectural flexibility (no invertibility required)
- ▶ Stable training (no adversarial objectives)
- ▶ Proven success in generative modeling

Neural Posterior Score Estimation (NPSE)

Forward Noising Process

Gradually add noise to transform $p(\theta|x)$ to tractable

$\pi(\theta) = \mathcal{N}(0, I)$:

$$d\theta_t = f(\theta_t, t)dt + g(t)dw_t$$

Backward Process

Time-reversal generates samples from posterior:

$$d\bar{\theta}_t = \left[-f(\bar{\theta}_t, T-t) + g^2(T-t)\nabla_{\theta} \log p_{T-t}(\bar{\theta}_t|x) \right] dt + g(T-t)dw_t$$

Key Insight

Only need score $\nabla_{\theta} \log p_t(\theta_t|x)$, not the density itself!

NPSE: Training and Sampling

Score Matching Objective

Train neural network $s_\psi(\theta_t, x, t) \approx \nabla_\theta \log p_t(\theta_t|x)$ by minimizing:

$$\mathcal{J}_{post}^{DSM}(\psi) = \frac{1}{2} \int_0^T \lambda_t \mathbb{E}_{p_{t|0}(\theta_t|\theta_0)p(x|\theta_0)p(\theta_0)} [\|s_\psi(\theta_t, x, t) - \nabla_{\theta_t} \log p_{t|0}(\theta_t|\theta_0)\|^2]$$

Sampling via Probability Flow ODE

$$\frac{d\theta_t}{dt} = \left[f(\theta_t, t) - \frac{1}{2} g^2(t) s_\psi(\theta_t, x_{obs}, t) \right]$$

Starting from $\theta_T \sim \mathcal{N}(0, I)$, integrate backward to obtain $\theta_0 \sim p(\theta|x_{obs})$

Visualization: Two Moons Example

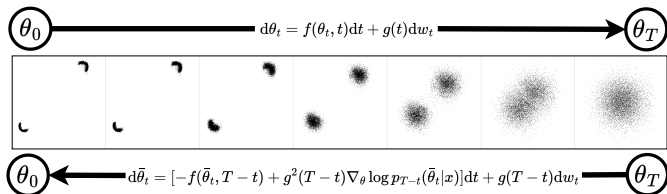


Figure: Forward process transforms posterior to Gaussian; backward process (with learned scores) generates posterior samples

Sequential Training: Motivation

Problem with Non-Sequential Approach

- ▶ Sample $\theta_i \sim p(\theta)$ uniformly from prior
- ▶ Most samples far from posterior at x_{obs}
- ▶ Inefficient use of limited simulation budget

Sequential Solution

- ▶ Adaptively guide simulations using current posterior approximation
- ▶ Focus computational resources on informative regions
- ▶ Use proposal distribution $\tilde{p}^r(\theta)$ that concentrates near x_{obs}

Central Challenge

How to correct for mismatch between proposal posterior and true posterior?

Truncated Sequential NPSE (TSNPSE)

Key Idea: Truncated Priors

In round r , define truncated prior using highest probability region:

$$\bar{p}^r(\theta) \propto p(\theta) \cdot \mathbb{I}\{\theta \in \text{HPR}_\epsilon(p_{\psi}^{r-1}(\theta|x_{\text{obs}}))\}$$

Proposal as Mixture

$$\tilde{p}^r(\theta) = \frac{1}{r} \sum_{s=0}^{r-1} \bar{p}^s(\theta)$$

where $\bar{p}^0(\theta) = p(\theta)$

Elegant Result

If we don't truncate regions with posterior support, then

$\tilde{p}^r(\theta) \propto p(\theta)$ within support.

No importance weight correction needed!

Alternative Sequential Approaches

SNPSE-A: Post-hoc Correction

- ▶ Generate samples from proposal posterior
- ▶ Resample using importance weights $w_i \propto p(\theta_i)/\tilde{p}^r(\theta_i)$
- ▶ Issue: Approximation error when learned proposal differs from true

SNPSE-B: Weighted Loss Function

- ▶ Include importance weights in loss: $\frac{p(\theta_0)}{\tilde{p}^r(\theta_0)} \parallel \cdot \parallel^2$
- ▶ Theoretically correct but high variance
- ▶ Unstable training

SNPSE-C: Score-space Correction

- ▶ Correct via score decomposition
- ▶ Requires multiple additional approximations

Benchmark Results: Non-Sequential Methods

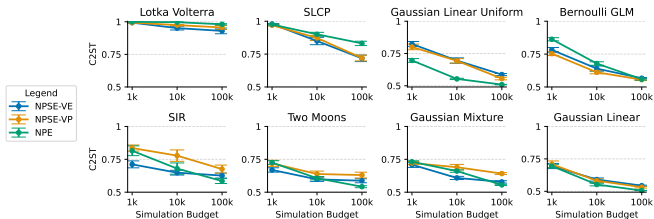


Figure: C2ST scores on 8 benchmarks (lower is better, 0.5 is perfect). NPSE competitive with NPE across tasks.

Benchmark Results: Sequential Methods

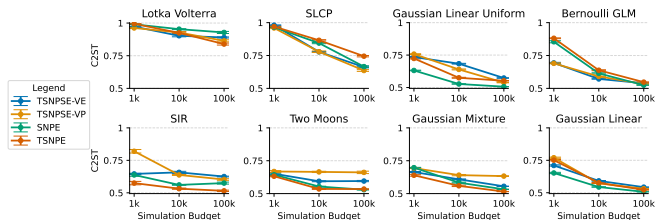


Figure: TSNPSE outperforms SNPE and TSNPE on challenging high-dimensional tasks (SLCP, Lotka-Volterra)

Real-World Application: Pyloric Network

Problem Characteristics

- ▶ 31 parameters governing neural dynamics
- ▶ 18 summary statistics from voltage traces
- ▶ >99% of prior samples produce invalid statistics
- ▶ Extremely challenging for SBI methods

TSNPSE Performance

- ▶ 9 rounds, 190k total simulations
- ▶ 81% valid statistics in final round (vs <60% for TSNPE)
- ▶ Posterior predictive matches observed data
- ▶ Marginals comparable to SNVI and TSNPE

Pyloric Results: Visualizations

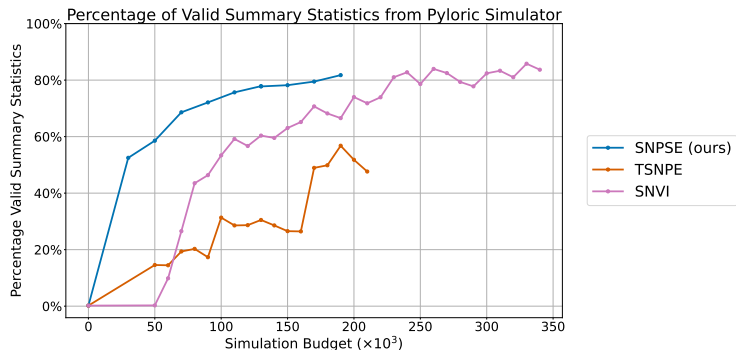


Figure: Top: Posterior mean predictive vs observation. Bottom: Percentage of valid statistics improves significantly with sequential focusing.

Conclusion

Key Contributions

- ▶ First application of score-based diffusion models to SBI
- ▶ TSNPSE: principled sequential training via truncated priors
- ▶ No importance weight corrections needed
- ▶ Architectural flexibility without normalizing flow constraints

Empirical Results

- ▶ Comparable or superior to SNPE on benchmarks
- ▶ Particularly strong on high-dimensional problems
- ▶ Successful on challenging real-world neuroscience application

Future Directions

Sequential variants of flow matching, specialized architectures for SBI, theoretical guarantees